Optical Flow Based Video Classification with visual attention using Recurrent Convolutional Neural Network

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Outline

- Introduction
- Background
  - Optical Flow
  - Convolutional Neural Networks
  - Recurrent Neural Network.
- Existing Methods on video classification.
- Proposed Method
- Results
- Conclusion and future work

Human Actions: Motivation

- Huge amount of video is available and growing
- Human actions are major events in movies, TV news, personal video …

BBC Motion Gallery

YouTube
Broadcast Yourself

150,000 uploads every day

Action recognition useful for:

- Content-based browsing

  e.g. fast-forward to the next goal scoring scene

- Video recycling

  e.g. find "Bush shaking hands with Putin"

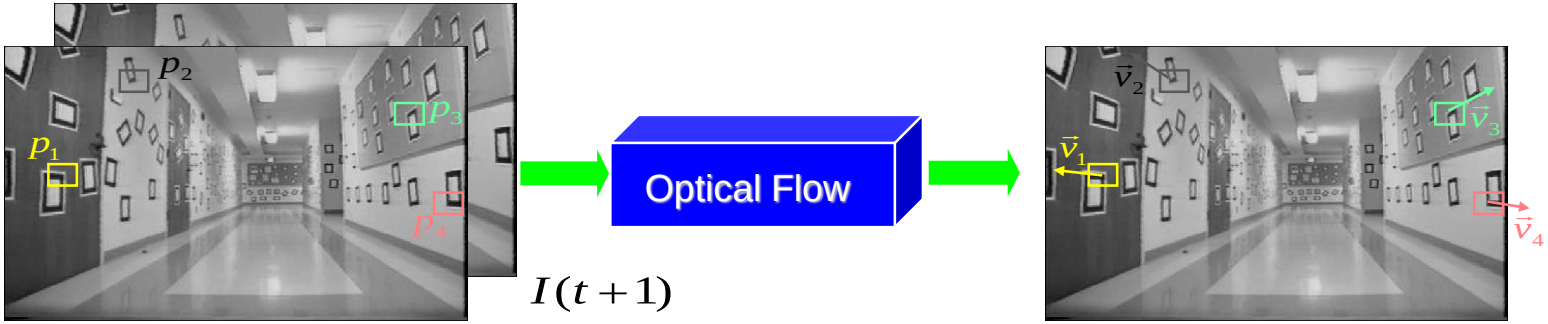
- Human scientists

  influence of smoking in movies on adolescent smoking

- Robotics

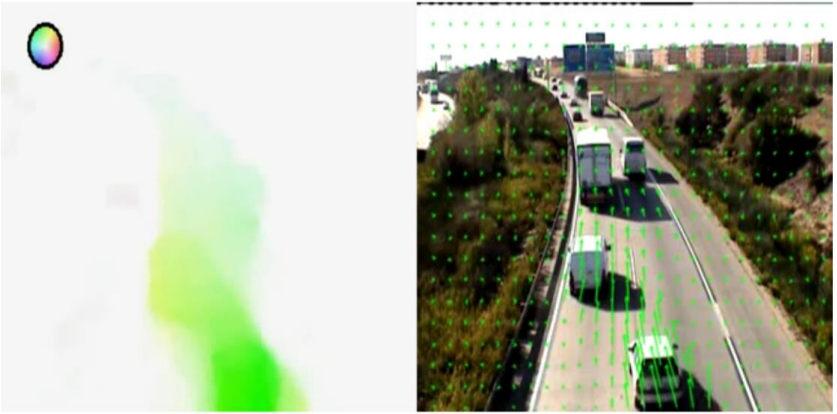
  e.g., Human-Robot Interaction

Optical Flow



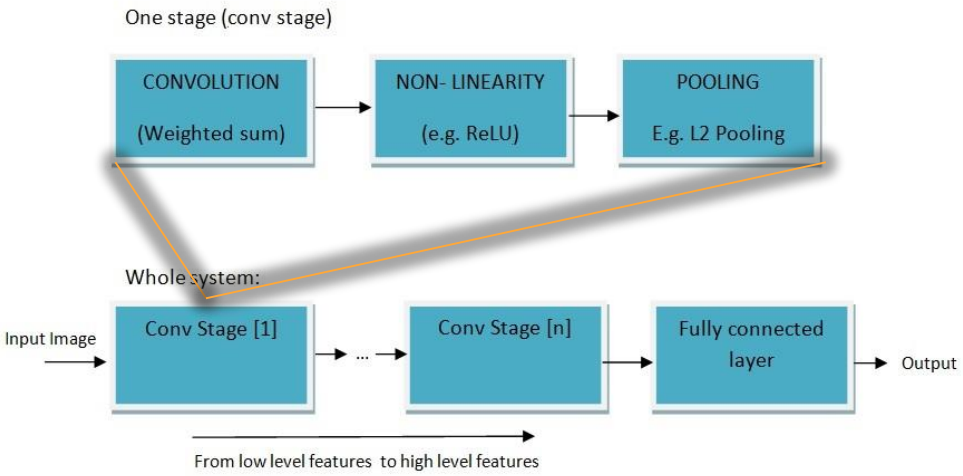
$I(t), \{p_i\}$

Velocity vectors $\{\vec{v}_i\}$

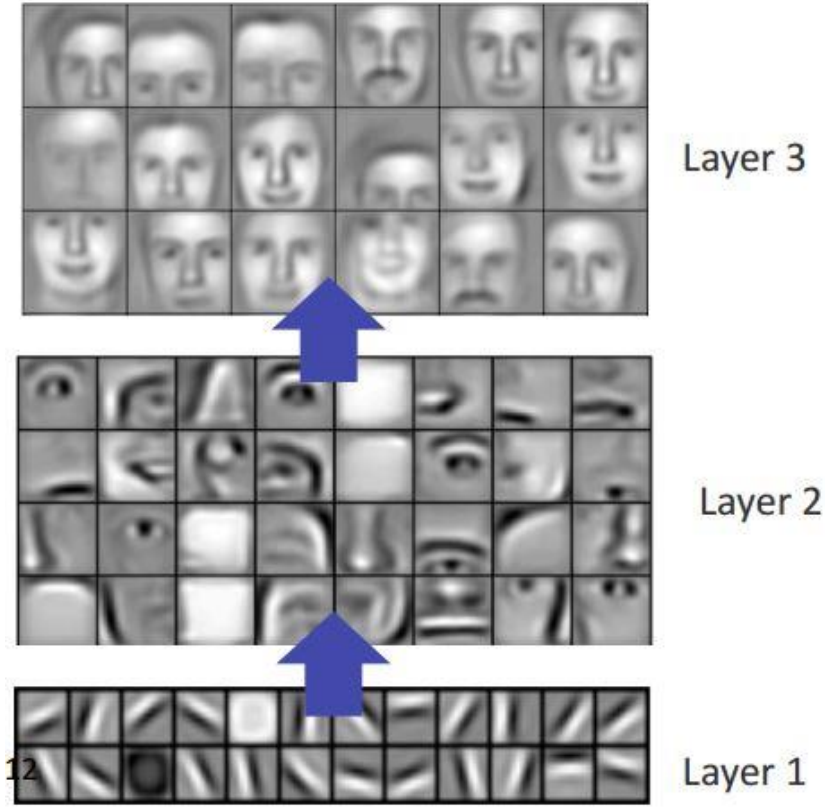


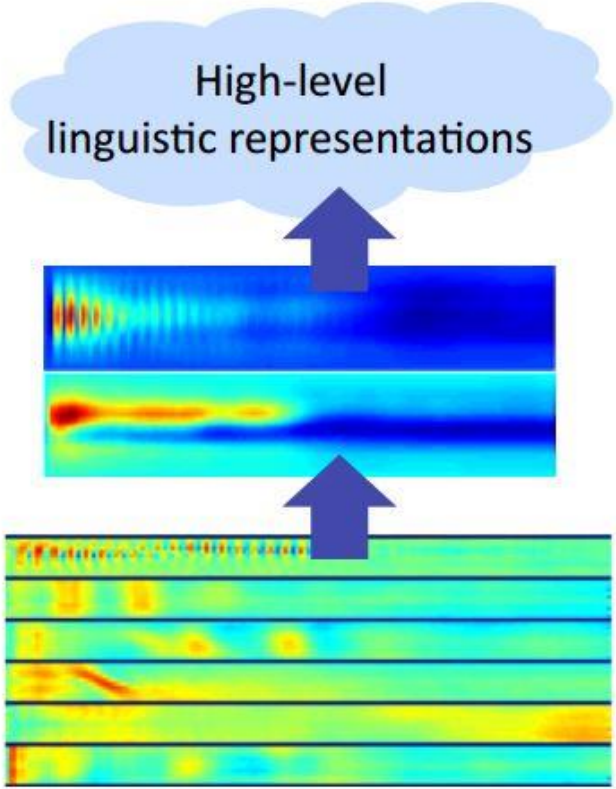
Convolutional Neural Networks (Convnets)

- Stacked stages of convolution, non-linearity, pooling followed by fully connected layers.
- The use of many layers is inspired from Mammalian visual cortex
- Reflects the structure of input
  - signals are compositional hierarchies



Convolutional Neural Networks (Convnets)





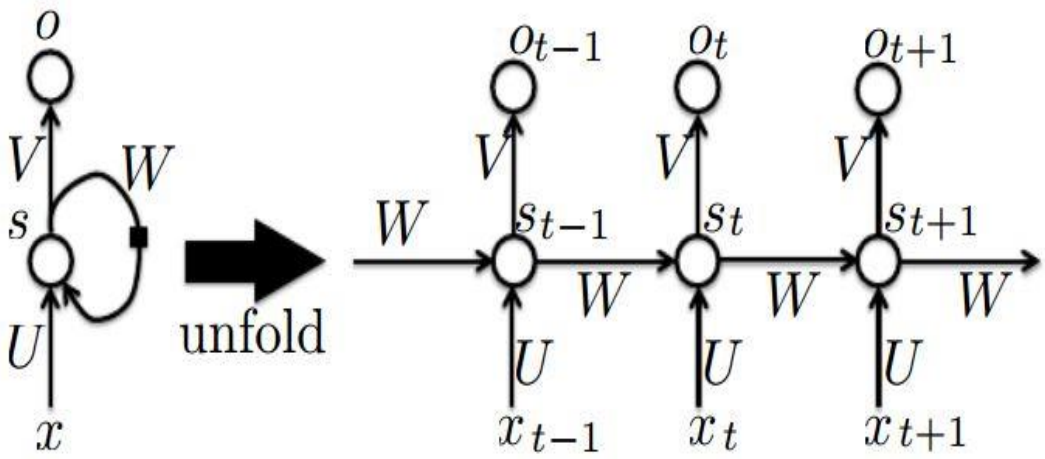
R.Socher, “Deep Learning for NLP (without magic)”, Stanford NLP-Report(2013)

Recurrent Neural Networks (RNN)

- Vanila RNN: hidden to hidden recurrence.
- Maps input sequence to output sequence of same length.
- Forward propagation equations:

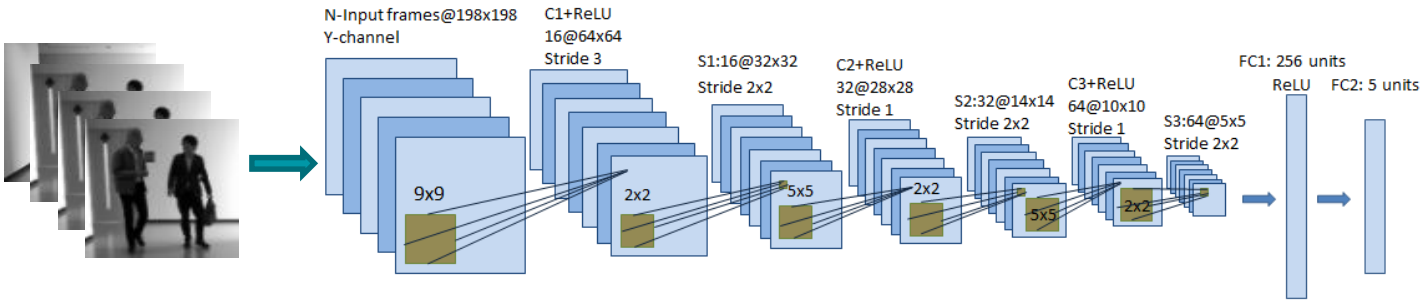
$$s_t = \tanh(b + Ws_{t-1} + Ux_t)$$

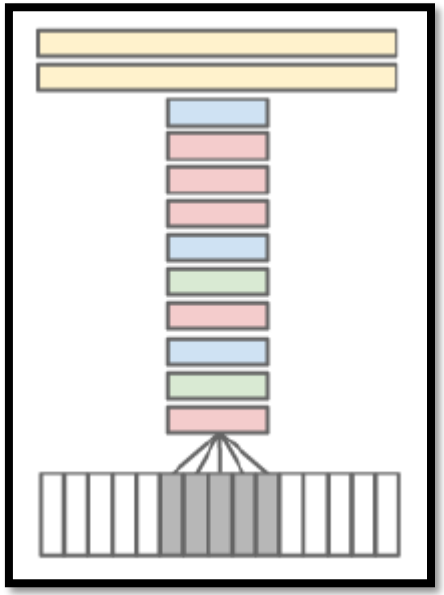
$$o_t = \mathrm{c} + \mathrm{V}\mathrm{s}_\mathrm{t}$$



Deep Learning by Yoshua Bengio, 2015

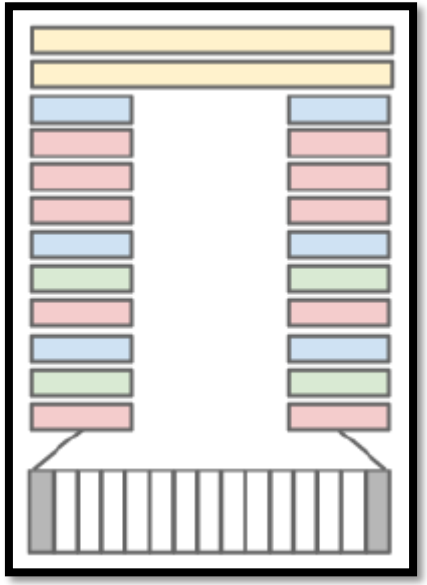
Early Fusion: ConvNets based Video Classifcation





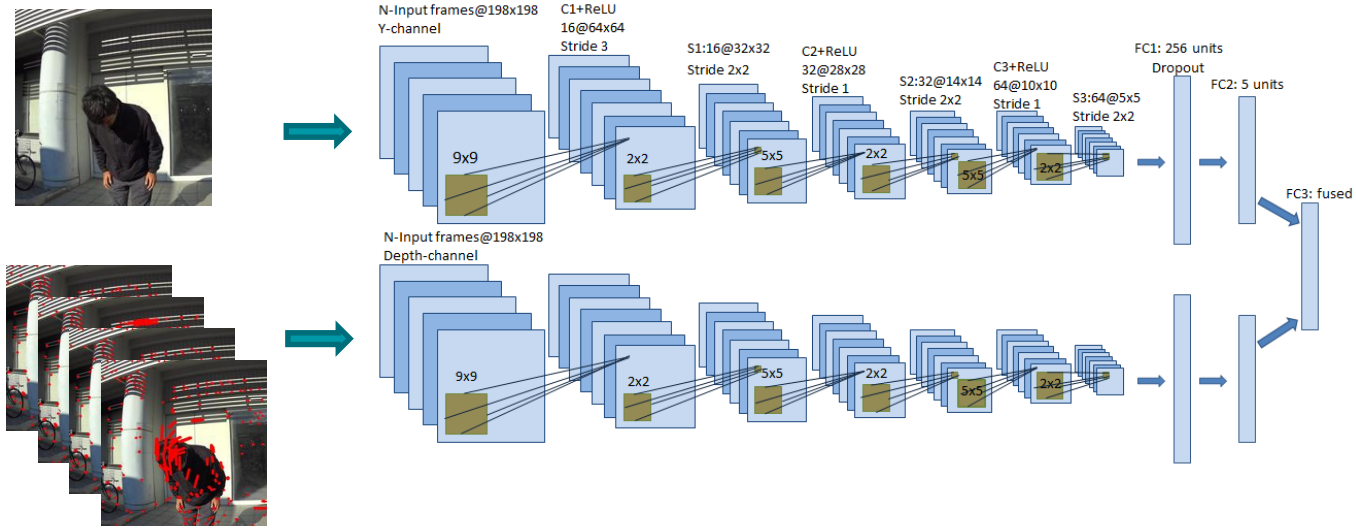
Karpathy, Andrej, et al. "Large-scale video classification with convolutional neural networks." *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*. 2014.

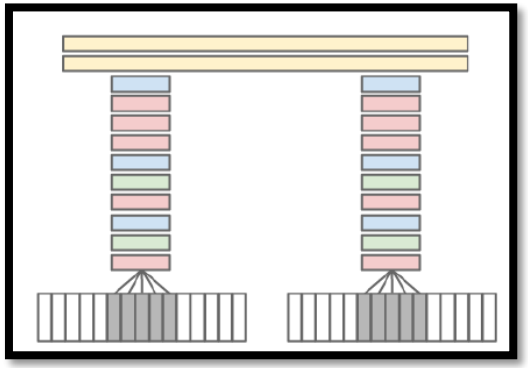
Late Fusion: ConvNets based Video Classifcation



Karpathy, Andrej, et al. "Large-scale video classification with convolutional neural networks." *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*. 2014.

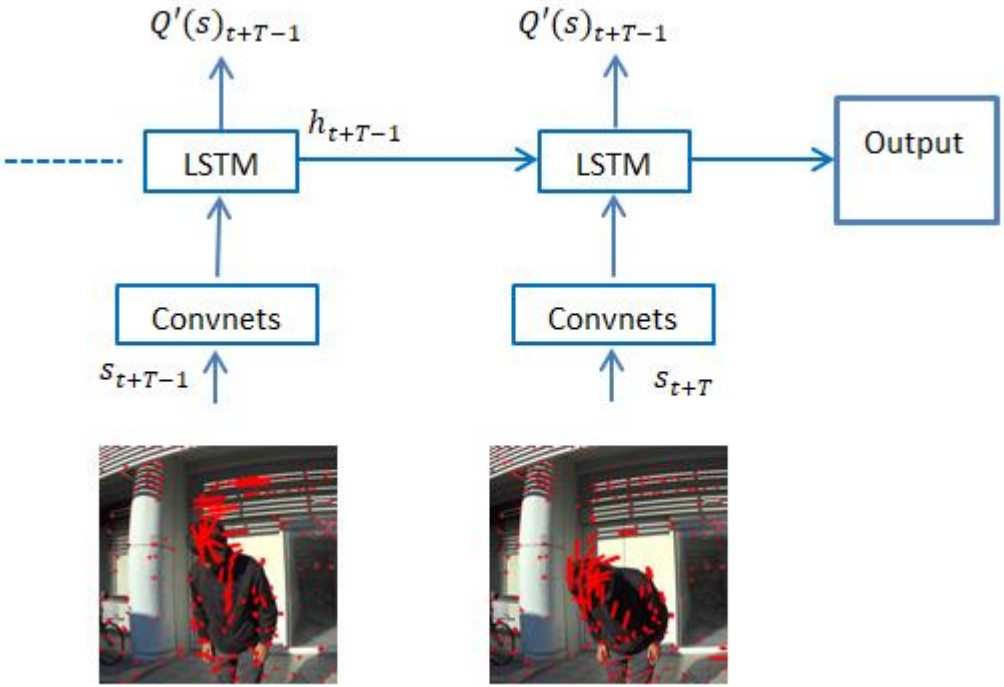
Two Streams: ConvNets based Video Classifcation

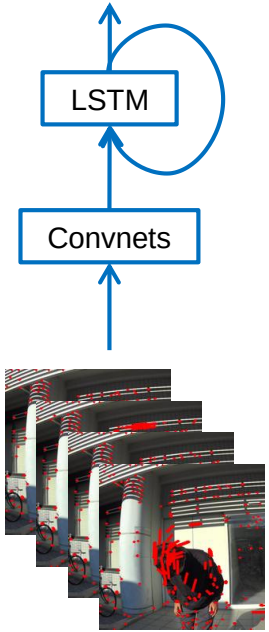




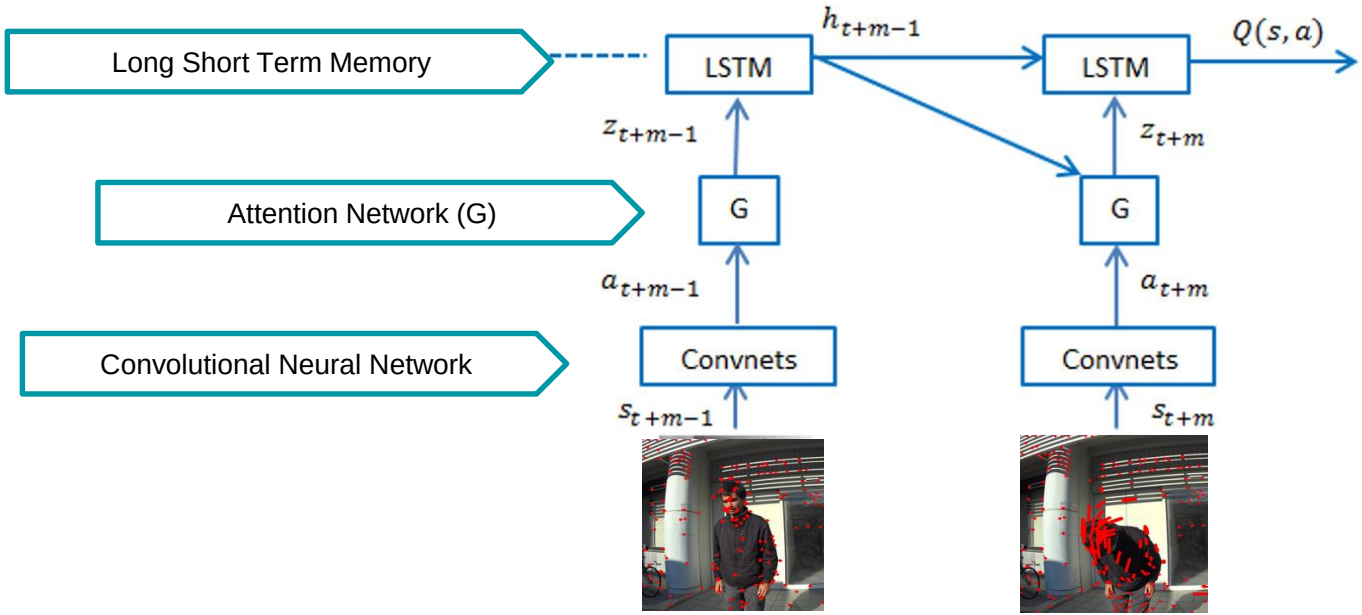
Simonyan, Karen, and Andrew Zisserman. "Two-stream convolutional networks for action recognition in videos." *Advances in neural information processing systems*. 2014.

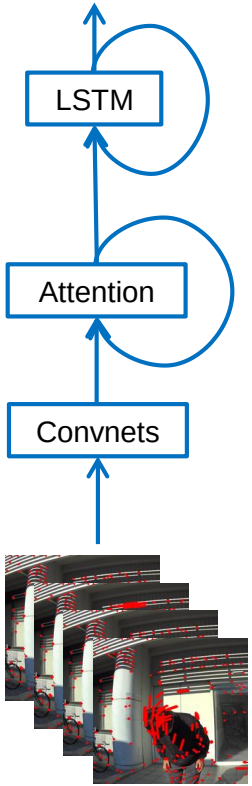
Recurrent ConvNets (R-CNN) based Video Classifcation





R-CNN with Visual Attention (R-ACNN) for Video Classifcation

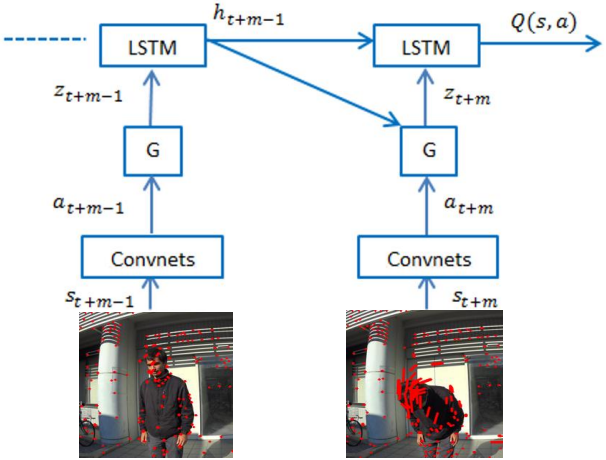


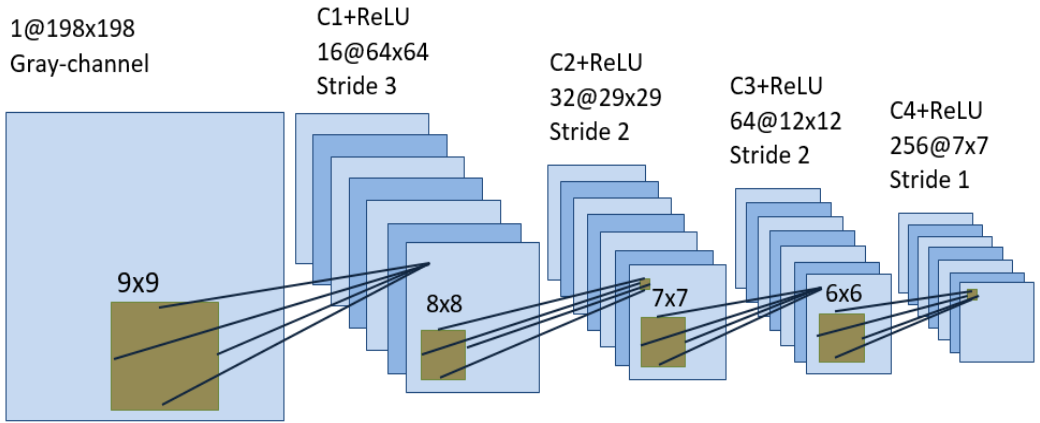


R-ACNN: Convents

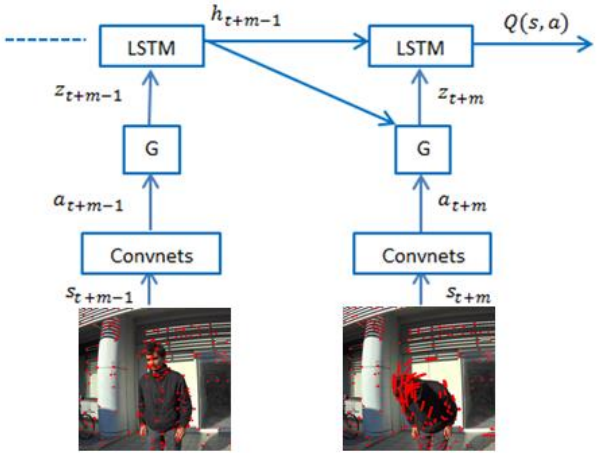
- Convnets takes an image of 198x198 as an input.
- Transform them to 256 feature maps of size 7x7 i.e.,

$$a_t = \{\boldsymbol{a}_t^1, \dots, \boldsymbol{a}_t^L\}; \boldsymbol{a}_t^l \in \mathbb{R}^D \text{ where } L = 256, D = 7\text{x}7$$





R-ACNN: Attention Network (G)



- Inputs:

  1) a_t: 256 vectors of dimension 7x7 from Convnets.
  2) h_{t-1}: previous hidden state of LSTM of dimension 256

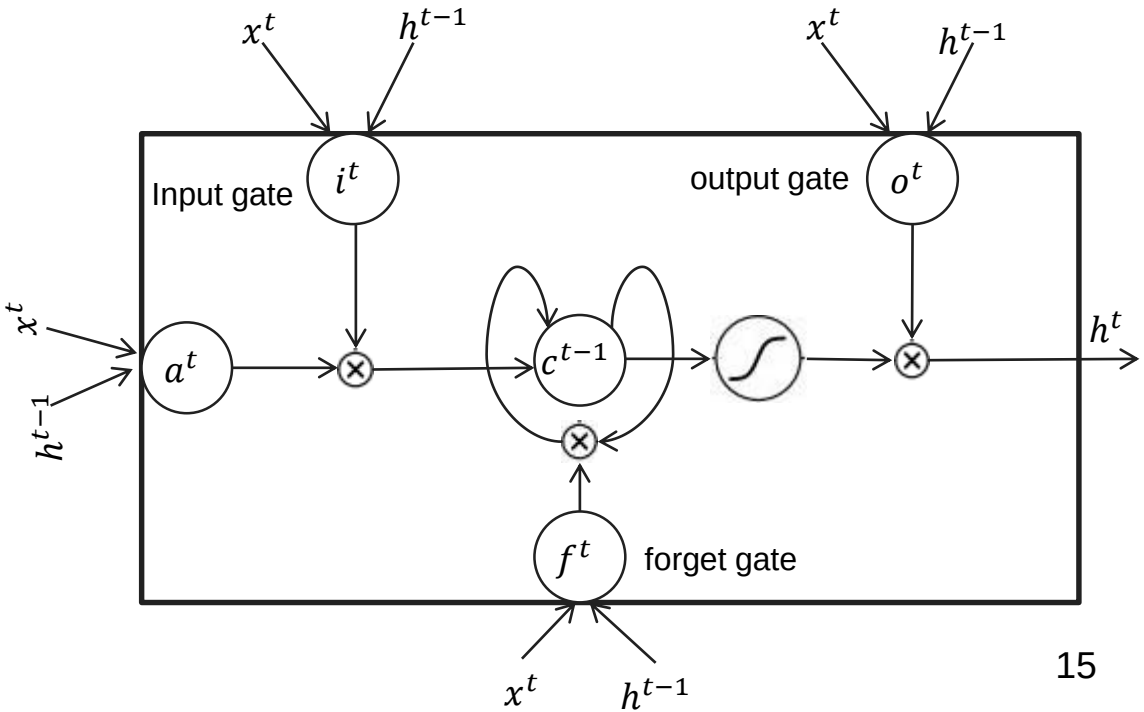
- Output:

  - Annotation vector $z_t = \sum_{l=1}^{L} \beta_t^l a_t^l$, where

$$\beta_t^l = \frac{\exp\left(W_1 \times \mathrm{Tanh}\left(W_2 \times a_t^l + W_3 \times h_{t-1}\right)\right)}{Z}$$

R-ACNN: LSTMs

- Input gate i^t: Controls what new information should be transferred to the cell c.
- Forget gate f^t: Controls how much accumulated information from past iterations should the cell keep.
- Output gate o^t: Controls what parts of the cell state should be transfer to the next state.
- The network is trained by back propagation.

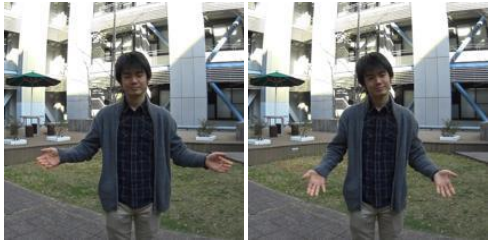


Training & Testing Dataset

1- Bowing (Konnichiwa)

2- Wave Hand (Hello)

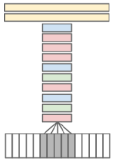
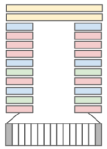
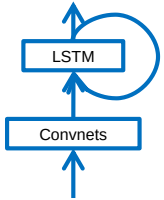
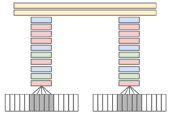
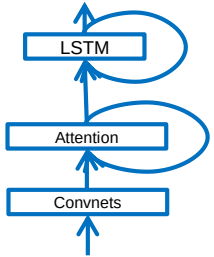
3- Don’t Know



4- Sshhh

5- Hi5

- 156 videos for each class.
  - Each of less than 2 sec
- Total: 780 videos
- Training:
  - 624 videos (80%)
- Testing:
  - 156 videos (20%)

Models	Early Fusion	Late Fusion	R-CNN	Two Stream	R-ACNN
Inputs					
Gray scale	42.8%	57.3%	59.0%	87.3%	57.8%
Optical Flow	-	**86.7%**	**98.2%**		**97.4%**

R-ACNN: Attention Mask

1- Bowing (Konnichiwa)

2- Don’t Know

R-ACNN: Attention Mask

3- Wave Hand (Hello)

4- Sshhh

5- Hi5

Recurrent Attentive Convolutional Neural Network

- Optical flow based video classification using R-ACNN
  - Provides visual attention
    - Precievability
    - Fewer training parameters compared to R-CNN/CNN
    - People show more willingness to interact with a system which can indicate attention.

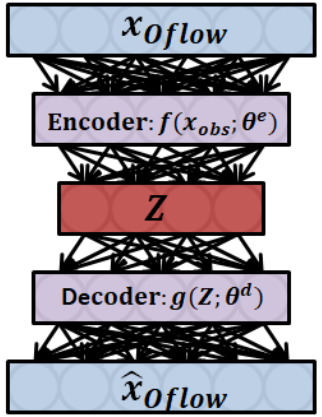
  - Significant improvement in prediction accuracy

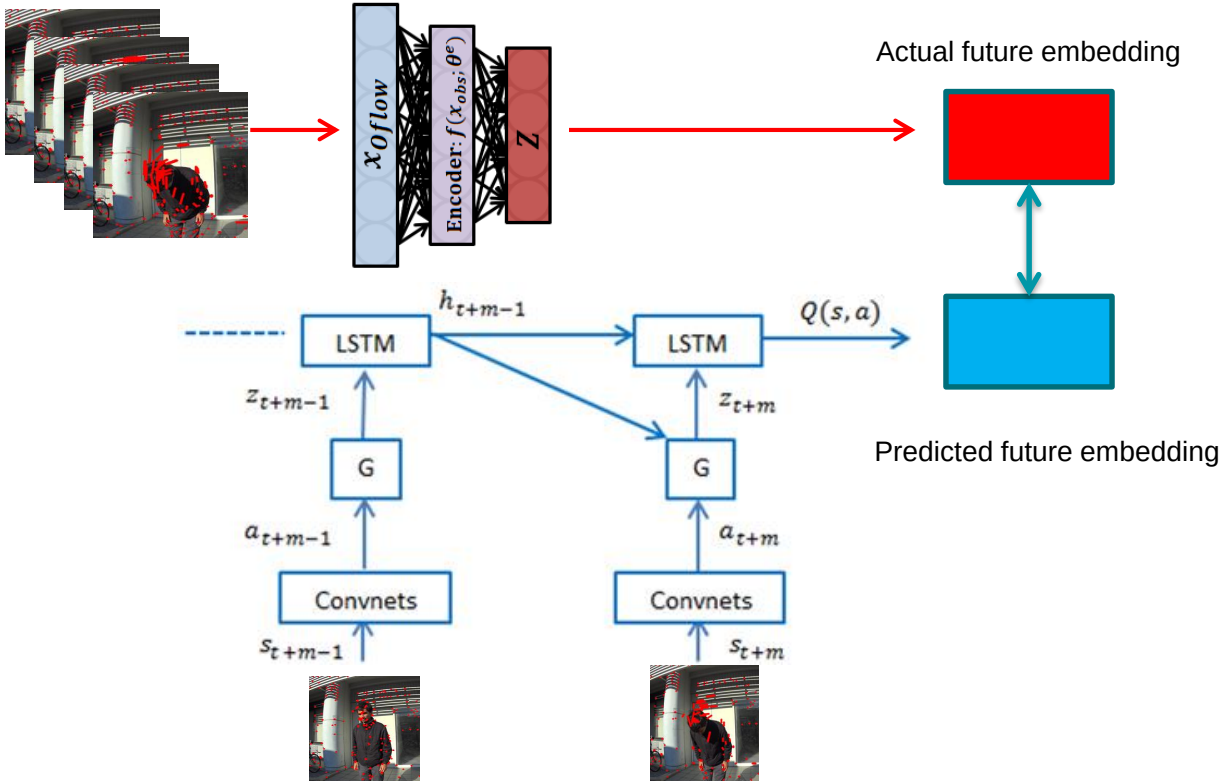
State-of-the-art [1]	R-ACNN/R-CNN
~89%	~97%

[1] Simonyan, Karen, and Andrew Zisserman. "Two-stream convolutional networks for action recognition in videos." *Advances in neural information processing systems*. 2014.

Unsupervised Optical Flow Prediction from Sequences

Optical Flow embedding





Thank you for your attention